

Queue Implementation
COP 3330

```
//Node.java
package qLecture;

/*
 * Node class in Linked List
 * COP3330
 * Author: Safa
 */
public class Node {

    private int item;
    private Node next;

    // we don't need any default node in our implementations
    // so we are stopping the developers calling no arg
    constructor
    private Node(){
        next=null;
        item=0;
    }

    public Node(int gitem) {
        item=gitem;
        next=null;
    }

    public Node getNext() {
        return next;
    }

    public int getItem() {
        return item;
    }

    public void setNext(Node n) {
        next=n;
    }

    private void setItem(int gitem) {
        item=gitem;
    }
}
```

```
//Queuex.java
package qLecture;
```

```
public interface Queuex {

    public void enqueue(int el);
    public int dequeue();
    public int peek();
    public boolean isEmpty();
}
```

```
//QueueL.java
package qLecture;
//LinkedList implementation of Queue
public class QueueL implements Queuex{
    Node head,tail;

    public QueueL() {
        head=null;
        tail=null;
    }

    public boolean isEmpty() {
        return head==null;
    }

    public void enqueue(int y) {
        Node n = new Node(y);
        if(!isEmpty()) {
            tail.setNext(n);
            tail=n;
        }else {
            head=n;
            //we need to set head also
            tail=n;
        }
    }

    public int dequeue() {
        int returned=-1;
        if(!isEmpty()) {
            returned=head.getItem();
            head=head.getNext(); //GC will take care of it
        }else {
            System.out.println("Empty list in dequeue");
        }
    }
}
```

```

        return returned;
    }

    public int peek() {
        int returned=-1;
        if(!isEmpty()) {
            returned=head.getItem();
        }else {
            System.out.println("Empty list in dequeue");
        }
        return returned;
    }
}

```

```

// QueueA.java
package qLecture;

// Array Implementation of Queue
// Written by Safa
// COP3330
public class QueueA implements QueueX{

    final int ASIZE=2;// array size
    int[] arr;
    int head,tail;

    QueueA(){
        head=0;
        tail=0;
        arr=new int[ASIZE];
    }

    @Override
    public void enqueue(int el) {
        if(tail<ASIZE) {
            arr[tail]=el;
            tail++;
        }else {
            System.out.println("Array Overflow");
        }
    }

    @Override
    public int dequeue() {
        int returned=-1;
        if(!isEmpty()) {

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```

        returned=arr[head];
        head++;
    }else {
        System.out.println("Empty Queue in Array Imp");
    }
    return returned;
}

@Override
public int peek() {
    int returned=-1;
    if(!isEmpty()) {
        returned=arr[head];
    }else {
        System.out.println("Empty Queue in Array Imp");
    }
    return returned;
}

@Override
public boolean isEmpty() {
    return head==tail;
}
}

```

```

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// QueueAL.java
package qLecture;
import java.util.ArrayList;
//ArrayList implementation
// Dequeue has run time complexity of O(n)
// which is not appropriate for standard Queue implementation

public class QueueAL implements Queueex {
    ArrayList<Integer> arr;

    QueueAL() {
        arr=new ArrayList<Integer>();
    }

    @Override
    public void enqueue(int el) {
        //Adding to the end of the list
        arr.add(el); // O(1)
    }
}

```

```

@Override
public int dequeue() {
    // TODO Auto-generated method stub
    if(isEmpty()) { //O(1)
        System.out.println("EMPTY QUEUE");
        return -1;
    }
    // removing the first element,
    //the arraylist shifts the remaining
    return arr.remove(0); // O(n)
}

@Override
public int peek() {
    if(isEmpty()) {
        System.out.println("EMPTY QUEUE");
        return -1;
    }
    return arr.get(0); //O(1)
}

@Override
public boolean isEmpty() {
    return arr.isEmpty();
}
}

```

```

// QueueTester.java
package qLecture;

// Queue Tester class
public class QueueTester {

    public static void main(String[] args) {
        QueueL ll=new QueueL();
        System.out.println("LINKEDLIST \n\n");
        qTest(ll);

        QueueA la=new QueueA();
        System.out.println("ARRAY \n\n");
        qTest(la);

        QueueAL arrL=new QueueAL();
        System.out.println("ARRAYLIST \n\n");
        qTest(arrL);

    }

    public static void qTest(Queue e) {
        e.enqueue(1);
        e.enqueue(2);
        e.enqueue(3);
        while(!e.isEmpty()) {
            System.out.println("Peek " + e.peek());
            System.out.println("Dequeue " + e.dequeue());
        }
    }

}

```